



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding with PyTorch Deep Learning

Course Code: C539

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	20 August 2026	Initial release. 4 topics, 21 hands-on labs, 2 days / 15 instructional hours.	Dr. Alfred Ang

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Course Information

Course Title	AI Vibe Coding with PyTorch Deep Learning
Course Reference	C539
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on practical labs (classroom, live online via Zoom, or on-site)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain what AI vibe coding is, set up Cursor, GitHub Copilot and Claude as PyTorch coding partners, and apply prompting patterns that produce correct deep learning code.
- LO2: Vibe code PyTorch tensor operations, computation graphs and autograd, and review and debug AI-generated PyTorch code rather than trusting it.
- LO3: Build regression and classification neural networks in PyTorch from prompts, choosing architectures, activation functions and loss functions deliberately.
- LO4: Generate training loops, optimizers and metrics from prompts, and save, load and iterate on trained models.
- LO5: Vibe code convolutional neural networks for image classification, diagnose overfitting, and apply data augmentation, regularization and transfer learning.
- LO6: Vibe code recurrent networks for sequence data, tune and evaluate them with follow-up prompts, and package a complete deep learning project.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Fundamentals to first networks - set up your AI pair programmer, vibe code tensors and autograd, then build, train and save regression and classification models.

Time	Duration	Topic / Activity
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9:30–9:50	20 min	Welcome, course introduction, ice-breaker and ground rules
9:50–10:25	35 min	Topic 1 — AI Vibe Coding for PyTorch Fundamentals (concepts + trainer demo)
10:25–11:15	50 min	Hands-on: Lab 1: What Is AI Vibe Coding — Your First PyTorch from a Prompt; Lab 2: Setting Up Cursor, GitHub Copilot and Claude for PyTorch
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 3: Prompting Patterns for Correct Deep Learning Code; Lab 4: Vibe Coding PyTorch Tensor Operations; Lab 5: Computation Graphs and Autograd with AI Assistance
13:00–14:00	60 min	Lunch break
14:00–14:20	20 min	Hands-on: Lab 6: Reviewing and Debugging AI-Generated PyTorch Code
14:20–14:55	35 min	Topic 2 — Vibe Coding Neural Networks (concepts + trainer demo)
14:55–15:45	50 min	Hands-on: Lab 7: Neural Network Architectures, Activation and Loss Functions; Lab 8: Vibe Coding a Regression Model in PyTorch
15:45–16:00	15 min	Tea break
16:00–17:50	110 min	Hands-on: Lab 9: Vibe Coding a Classification Model with Softmax and Cross Entropy; Lab 10: Generating Training Loops, Optimizers and Metrics from Prompts; Lab 11: Saving, Loading and Iterating on Models
17:50–18:15	25 min	Consolidation for Topic 1 and Topic 2: reading, predicting and correcting AI-generated PyTorch — shapes, losses and the training loop
18:15–18:30	15 min	Day 1 recap and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — Vision and sequences - vibe code a CNN image classifier, beat overfitting with augmentation and transfer learning, then forecast with an LSTM and package the project.

Time	Duration	Topic / Activity
9:30–9:40	10 min	Day 1 review, outstanding questions and objectives for the day

9:40–10:15	35 min	Topic 3 — Vibe Coding Convolutional Neural Networks (concepts + trainer demo)
10:15–11:15	60 min	Hands-on: Lab 12: Overview of CNNs: Convolution, Pooling and Padding; Lab 13: Vibe Coding a CNN Image Classifier
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 14: Diagnosing Overfitting with AI Assistance; Lab 15: Data Augmentation and Regularization via Prompts; Lab 16: Transfer Learning with Pre-Trained Models
13:00–14:00	60 min	Lunch break
14:00–14:35	35 min	Topic 4 — Vibe Coding Recurrent Networks for Sequence Data (concepts + trainer demo)
14:35–15:45	70 min	Hands-on: Lab 17: Overview of RNNs, LSTM and GRU; Lab 18: Vibe Coding an LSTM for Time Series Forecasting
15:45–16:00	15 min	Tea break
16:00–17:50	110 min	Hands-on: Lab 19: Tuning Sequence Models with Follow-Up Prompts; Lab 20: Evaluating and Visualizing Model Performance; Lab 21: Packaging a Complete Deep Learning Project
17:50–18:15	25 min	Project showcase: learners demo the packaged ForgeSight project and walk through the prompts that built each model
18:15–18:30	15 min	Course recap, Q&A, next steps and close

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: AI Vibe Coding for PyTorch Fundamentals	~29% of course time	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6
Topic 02: Vibe Coding Neural Networks	~24% of course time	Lab 7, Lab 8, Lab 9, Lab 10, Lab 11
Topic 03: Vibe Coding Convolutional Neural Networks	~24% of course time	Lab 12, Lab 13, Lab 14, Lab 15, Lab 16
Topic 04: Vibe Coding Recurrent Networks for Sequence Data	~23% of course time	Lab 17, Lab 18, Lab 19, Lab 20, Lab 21